

CPGSV17 Blocked-Basis Chi Families and Uniform BNT Labels

2026-05-24

1 Source statement

Cirac–Pérez–García–Schuch–Verstraete 2017 state in [CPGSV17, Theorem IV.13(ii)] that there is a set of diagonal matrices

$$\chi_{\alpha,\beta,\gamma}$$

with positive entries such that, for every length L , the operators $O_L(M_\alpha)$ linearly span an algebra whose structure coefficients satisfy

$$c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L).$$

The important point for the present note is that the labels α, β, γ are BNT labels, not basis labels depending on L . Thus the same diagonal matrix $\chi_{\alpha,\beta,\gamma}$ is used for all positive lengths; only the exponent changes.

Here $O_L(M_\alpha)$ denotes the length- L operator associated to the normal tensor labelled by α in the basis of normal tensors. The phrase “BNT labels” refers to these labels α, β, γ . The support algebra $\mathcal{A}_n$ below is the formal support algebra chosen for the blocked length n in the Lean development.

2 Formalized blocked-basis analogue

The current blocked coefficient construction in `TNLean/MPS/MPD0/AlgebraStructure.lean` chooses bases of the support algebras $\mathcal{A}_n$. Its multiplication coefficients have indices

$$i, j \in \text{Basis}(\mathcal{A}_n), \quad k \in \text{Basis}(\mathcal{A}_{2n}),$$

because multiplication is represented as

$$\mathcal{A}_n \times \mathcal{A}_n \longrightarrow \mathcal{A}_{2n}.$$

The declarations `MPOTensor.AlgebraStructureData.BlockedStructureChiFamily` and `MPOTensor.AlgebraStructureData.HasBlockedStructureChiTracePowerForm`

therefore express a positive-length, length-dependent blocked-basis analogue:

$$c_{i,j,k}^{(n)} = \sum_r \chi_{n,i,j,k,r}^n = \text{tr}(\chi_{n,i,j,k}^n).$$

Here n is the source blocking length of the two factors in $\mathcal{A}_n$, while the coefficient k is taken in the basis of $\mathcal{A}_{2n}$. The predicate is intentionally restricted to $0 < n$, avoiding the degenerate length-zero constraint $c_{i,j,k}^{(0)} = \dim(\chi_{0,i,j,k})$, which is not part of the intended positive-chain statement.

3 Gap

Verdict: scope restriction.

This blocked-basis analogue is not the uniform BNT-label statement of Theorem IV.13(ii). It allows the diagonal matrix to depend on n , because the available indices themselves depend on the blocked bases $\mathcal{A}_n$ and $\mathcal{A}_{2n}$. The source theorem instead provides one matrix for each triple of BNT labels and uses that same matrix for every length L .

There is a second scope restriction in the operation being compared. In the paper, the coefficients $c_{\alpha,\beta,\gamma}^{(L)}$ belong to the same-length operator algebra:

$$O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma c_{\alpha,\beta,\gamma}^{(L)} O_L(M_\gamma).$$

The current Lean coefficient family comes from the blocked support-algebra product

$$\mathcal{A}_n \times \mathcal{A}_n \longrightarrow \mathcal{A}_{2n},$$

so the two input factors have source length n , while the result is expanded in a basis at length $2n$. The exponent n in

$$c_{i,j,k}^{(n)} = \text{tr}(\chi_{n,i,j,k}^n)$$

is therefore the source length of the two factors, not the codomain length. This convention is only a blocked-basis analogue; it is not a replacement for the same-length multiplication statement in Theorem IV.13(ii).

This note isolates the chi-uniformity and same-length coefficient comparison part of Theorem IV.13(ii). The idempotent condition

$$m_\gamma = \sum_{\alpha,\beta} c_{\alpha,\beta,\gamma}^{(1)} m_\alpha m_\beta$$

from paper lines 981–985 is a separate condition in the same theorem; it is not represented by the blocked-basis trace-power predicate.

Consequently, `MPOTensor.AlgebraStructureData.HasBlockedStructureChiTracePowerForm` should not be cited as a formalization of Theorem IV.13(ii). It is a local blocked-basis coefficient statement, useful only after the uniform BNT-label statement has been separated from the chosen blocked bases.

4 Relation to the algebra-to-fusion converse

Let $E = E_1$ be the one-site transfer map. The present algebra predicate gives the fixed-point-space identity

$$\text{Fix}((E^n)^\dagger) = \text{Fix}(E^\dagger), \quad n > 0.$$

This rules out root-of-unity peripheral eigenvalues of $E^\dagger$, but it does not imply the fusion-side idempotence equation $E^2 = E$.

The converse direction (ii) $\Rightarrow$ (i) in the proof of Theorem IV.13 uses a stronger coefficient argument. In Appendix C.4, lines 2046–2085, the authors compare the two BNT expansions coming from the one-site and two-site decompositions. The source identity `eq:algebra-appendix` is stated at lines 1933–1936 and used in this converse at lines 2048–2050:

$$\sum_{\gamma} \text{tr}(\nu_{\gamma}^L) O_L(A_{\gamma}) = \sum_{\alpha, \beta, \gamma} \text{tr}((\mu_{\alpha} \otimes \mu_{\beta} \otimes \chi_{\alpha, \beta, \gamma})^L) O_L(M_{\gamma}).$$

Since both A_{γ} and M_{γ} are BNT, the comparison first gives, up to relabelling (line 2053),

$$A_{\gamma} = e^{i\theta_{\gamma}} Z_{\gamma} M_{\gamma} Z_{\gamma}^{-1}.$$

Proposition IV.12 then removes the phase and identifies Z_{γ} with a unitary U_{γ} , as in line 2057, so

$$A_{\gamma} = U_{\gamma} M_{\gamma} U_{\gamma}^{-1}$$

with U_{γ} unitary. Since, for all sufficiently large L , the operators $O_L(A_{\gamma}) = O_L(M_{\gamma})$ are linearly independent, and all coefficients are positive, the simple-matrix lemma (lines 1155–1163, used at line 2058) gives the multiset identity

$$\{\nu_{\gamma, k}\}_k = \{\mu_{\alpha, k_1} \mu_{\beta, k_2} \chi_{\alpha, \beta, \gamma, k_3}\}_{\alpha, \beta, k_1, k_2, k_3}.$$

Summing this identity gives

$$\text{tr}(\nu_{\gamma}) = \sum_{\alpha, \beta} m_{\alpha} m_{\beta} c_{\alpha, \beta, \gamma}^{(1)} = m_{\gamma}.$$

Only after this trace equality is obtained do the normalizing maps in [CPGSV17, lines 2065–2085] have the form

$$\tilde{R}_2\left(\bigoplus_{\gamma} X_{\gamma}\right) = \bigoplus_{\gamma} \frac{\nu_{\gamma}}{m_{\gamma}} \otimes X_{\gamma}, \quad \tilde{R}_1\left(\bigoplus_{\gamma} X_{\gamma}\right) = \bigoplus_{\gamma} \frac{\mu_{\gamma}}{m_{\gamma}} \otimes X_{\gamma}.$$

These are the maps used in $T = \tilde{R}_2 \tilde{T} R_1$ and $S = \tilde{R}_1 \tilde{S} R_2$. The first denominator is normalized by $\text{tr}(\nu_{\gamma}) = m_{\gamma}$, proved above; the second by the defining equality $m_{\gamma} = \text{tr}(\mu_{\gamma})$ for the original block coefficient.

The present algebra predicate `MPOTensor.isRFP`MPDO`via`algebra` does not include these BNT-label coefficients, the positive $\chi_{\alpha, \beta, \gamma}$, or the length-one idempotent trace identity. It proves the fixed-point-space equation above, not the displayed BNT coefficient comparison. Thus the source-faithful route to the fusion formulation must construct the BNT-label theorem witness from the MPDO tensor and then use the one-site retract criterion `MPOTensor.fusionIsometryData`one`iff`isRFP`.

5 BNT-label coefficient objects and remaining elimination plan

The same-length coefficient family $c_{\alpha,\beta,\gamma}^{(L)}$, indexed by fixed BNT labels, is represented separately from the blocked-basis coefficients.¹ When the Appendix C.4 construction has already produced a diagonal $\chi_{\alpha,\beta,\gamma}$ -family, there is also a canonical coefficient family

$$c_{\alpha,\beta,\gamma}^{(L)} = \sum_r \chi_{\alpha,\beta,\gamma,r}^L.$$

This is recorded separately so that the χ -family can serve as the single source of those coefficients in the source-side special case.² A same-length BNT product predicate records the algebra identity

$$O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma c_{\alpha,\beta,\gamma}^{(L)} O_L(M_\gamma)$$

for an abstract family of length- L BNT operators.³ The idempotent condition

$$m_\gamma = \sum_{\alpha,\beta} c_{\alpha,\beta,\gamma}^{(1)} m_\alpha m_\beta$$

is also represented as a predicate on the trace scalars $m_\alpha = \text{tr}(\mu_\alpha)$.⁴ A positive BNT-label χ -witness consists of a length-independent diagonal family $\chi_{\alpha,\beta,\gamma}$, positivity of every diagonal entry, and the positive-length trace-power identity.⁵ If the coefficients are chosen canonically from the same χ -family, then positivity of the diagonal entries alone gives such a witness.⁶ A blocked-basis comparison predicate records that the chosen blocked multiplication coefficients are the pullback of the source-length BNT-label coefficients along a source label map from $\mathcal{A}_n$ and a target label map from $\mathcal{A}_{2n}$. The label-assignment datum is separated from the coefficient equality, so the Appendix C.3 construction target can be stated without also assuming the comparison formula.⁷ This predicate inherits the second scope restriction above: its left hand side is expanded in a basis of $\mathcal{A}_{2n}$, while the source theorem's BNT product law is same-length. It therefore records the formal blocked-basis comparison predicate; constructing the source label maps from an MPDO tensor is still part of the Appendix C.3–C.4 witness construction. The predicate is not itself the same-length product

¹ `MPOTensor.BNTLabelCoefficientFamily`.

² `MPOTensor.BNTLabelCoefficientFamily.ofChi`, `MPOTensor.BNTLabelCoefficientFamily.ofChi'coeff`, `MPOTensor.BNTLabelCoefficientFamily.ofChi'coeff'eq'trace'matrix'pow`, `MPOTensor.BNTLabelCoefficientFamily.ofChi'hasPositiveLengthChiTracePowerForm`.

³ `MPOTensor.BNTLabelOperatorFamily.HasSameLengthProductForm`.

⁴ `MPOTensor.BNTLabelTraceScalarFamily.HasIdempotentCoefficientForm`.

⁵ `MPOTensor.PositiveBNTLabelChiTracePowerForm`.

⁶ `MPOTensor.PositiveBNTLabelChiTracePowerForm.ofChi`.

⁷ `MPOTensor.BNTBlockedBasisLabelAssignment`, `MPOTensor.BNTBlockedBasisLabelAssignment.blockedLabel`, `MPOTensor.BNTBlockedBasisCoefficientComparison`.

statement of Theorem IV.13(ii). These declarations specify the BNT-label coefficient objects and the comparison statement, but they do not yet construct the operators, trace scalars, coefficients, comparison maps, or χ -matrices from an MPDO tensor. Once both the same-length BNT product form and a positive BNT-label χ -witness are available, the same-length product expansion with coefficients written as $\text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ is formalized as a direct corollary.⁸ Once both the comparison predicate and a positive BNT-label χ -witness are available, the coefficient-level blocked trace-power formula is formalized as a direct corollary.⁹ The same hypotheses also give a positive blocked-basis χ -witness by pulling back the uniform BNT-label χ -family along the comparison maps.¹⁰ Likewise, once the idempotent coefficient condition and the positive BNT-label χ -witness are available, the length-one idempotent identity is formalized with the coefficients rewritten as traces of the corresponding χ -matrices.¹¹

The BNT-label objects above are now also gathered into one theorem-data record: fixed coefficients, same-length operators and product law, trace scalars and idempotent condition, positive BNT-label χ -witness, and blocked-basis comparison.¹² This record is not an additional mathematical assumption in the source theorem; it is the single formal object collecting the source data supplied by the Appendix C.3–C.4 construction from an MPDO tensor. There is also a construction for the source-side special case in which the coefficient family is the one canonically determined by the same χ -family; it takes the product law, idempotent law, and blocked-basis comparison for those canonical coefficients.¹³ From such data, the source predicates themselves and the corresponding equation-level consequences are available: the same-length product law, the idempotent scalar law, the BNT-label χ -positivity and trace-power predicates, the coefficient trace-power formula, positivity of the diagonal χ -entries, the same-length chi-trace product law, the chi-trace idempotent law, the blocked-basis coefficient trace-power formula, the blocked-basis coefficient comparison, the positive blocked χ -witness, the pulled-back blocked trace-power predicate, and the pulled-back blocked-basis χ -family with positive entries and trace-power formula are immediate consequences.¹⁴

⁸ `MPOTensor.BNTLabelOperatorFamily.HasSameLengthProductForm.eq'sum'chi'trace'pow,`
`MPOTensor.BNTLabelOperatorFamily.HasSameLengthProductForm.eq'sum'ofChi'trace'pow.`

⁹ `MPOTensor.BNTBlockedBasisCoefficientComparison.blocked'coeff'eq'trace'pow,` `MPOTensor.BNTBlockedBasisCoefficientComparison.blocked'coeff'eq'ofChi'trace'pow,` `MPOTensor.BNTBlockedBasisCoefficientComparison.blocked'coeff'eq'pulledBlockedChi'trace'pow.`

¹⁰ `MPOTensor.BNTBlockedBasisCoefficientComparison.pulledBlockedChiFamily'toDiagonal'of'pos,`
`MPOTensor.BNTBlockedBasisCoefficientComparison.pulledBlockedChi'tracePowerCoeff'of'pos,`
`MPOTensor.BNTBlockedBasisCoefficientComparison.pulledBlockedChi'dim'of'pos,` `MPOTensor.BNTBlockedBasisCoefficientComparison.pulledBlockedChi'trace'matrix'pow'of'pos,` `MPOTensor.BNTBlockedBasisCoefficientComparison.toPositiveBlockedStructureChiTracePowerForm.`

¹¹ `MPOTensor.BNTLabelTraceScalarFamily.HasIdempotentCoefficientForm.eq'sum'chi'trace,`
`MPOTensor.BNTLabelTraceScalarFamily.HasIdempotentCoefficientForm.eq'sum'ofChi'trace.`

¹² `MPOTensor.BNTLabelTheoremData.`

¹³ `MPOTensor.BNTLabelTheoremData.ofChi.`

¹⁴ `MPOTensor.BNTLabelTheoremData.same'length'product'form,` `MPOTensor.BNTLabelTheoremData.idempotent'coefficient'form,` `MPOTensor.BNTLabelTheoremData.positive'chi'pos'entries,` `MPOTensor.BNTLabelTheoremData.positive'chi'trace'power,` `MPOTensor.BNTLabelTheoremData.labelAssignment,` `MPOTensor.BNTLabelTheoremData.labelAssignment,`

The same data can be stated in the existential form closest to the paper: one object records the BNT-label type, the same-length operator spaces, their algebraic structure, and the corresponding theorem data.¹⁵ The existential nonempty statement can also be unpacked directly into a concrete witness carrying the source predicates, the corresponding same-length and idempotent equations written with $c_{\alpha,\beta,\gamma}^{(L)}$, and the same two equations with coefficients written as χ -traces; the blocked-basis χ -family is also identified as the pullback of the BNT-label χ -family along the comparison maps.¹⁶ The existential statement has the analogous canonical construction from a positive χ -family, again leaving the product, idempotent, and blocked-comparison statements as inputs rather

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15MPOTensor.BNTLabelTheoremWitness,      MPOTensor.HasBNTLabelTheoremWitness,
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16MPOTensor.HasBNTLabelTheoremWitness.exists'source'predicates,      MPOTen-
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sor.HasBNTLabelTheoremWitness.exists'source'chi'trace'equations.

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than reconstructing them from an MPDO tensor.¹⁷ Such a witness immediately gives the source predicates and their consequences: the coefficient trace-power formula, the same-length product law, the idempotent scalar law, positivity of the diagonal χ -entries, the blocked-basis coefficient comparison, the same-length chi-trace product law, the chi-trace idempotent law, the blocked-basis coefficient trace-power formula, the positive blocked-basis χ -witness, and the pulled-back blocked trace-power predicate and χ -family for the chosen algebra-structure data.¹⁸ The remaining step is still the construction of this existential witness from the MPDO tensor.

To eliminate the remaining gap, the formalization should introduce:

1. construction of the comparison maps relating the BNT-label coefficients to the chosen blocked bases of $\mathcal{A}_n$;
2. construction of the same-length BNT operator family and product law from the MPDO tensor, together with the comparison to the blocked support-algebra product $\mathcal{A}_n \times \mathcal{A}_n \rightarrow \mathcal{A}_{2n}$;
3. construction of the positive BNT-label $\chi_{\alpha,\beta,\gamma}$ -witness from the paper's Appendix C.3 and C.4 hypotheses;

¹⁷MPOTensor.BNTLabelTheoremWitness.ofChi, MPOTensor.BNTLabelTheoremWitness.hasBNTLabelTheoremWitness, MPOTensor.BNTLabelTheoremWitness.ofChi'hasBNTLabelTheoremWitness.

¹⁸MPOTensor.BNTLabelTheoremWitness.same'length'product'form,
MPOTensor.BNTLabelTheoremWitness.idempotent'coefficient'form,
MPOTensor.BNTLabelTheoremWitness.positive'chi'pos'entries, MPOTen-
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sor.BNTLabelTheoremWitness.positiveBlockedChi'entry'pos, MPOTen-
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4. construction of the idempotent condition $m_\gamma = \sum_{\alpha,\beta} c_{\alpha,\beta,\gamma}^{(1)} m_\alpha m_\beta$ from the MPDO tensor;
5. construction of the single `MPOTensor.BNTLabelTheoremWitness` from these source objects, and hence of the underlying `MPOTensor.BNTLabelTheoremData` record. The blocked-basis trace-power statement, with its source-length exponent convention, is then a derived corollary of the uniform BNT-label theorem, rather than a separate replacement for the theorem itself.

References

- [CPGSV17] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. *Annals of Physics*, 378:100–149, 2017.